Formula Book: Algorithms and Computational Complexity

1 Big O Notation

Let $f, g : \mathbb{N} \to \mathbb{R}_{\geq 0}$, with $g(n) > 0$ for all sufficiently large n. We say that

$$f(n) = O(g(n))$$

if there exist constants $c > 0$ and $n_0 \in \mathbb{N}$ such that

$$0 \leq f(n) \leq c\, g(n) \qquad \text{for all } n \geq n_0.$$

Thus, $g(n)$ is an asymptotic upper bound for $f(n)$. If the following limit exists, then

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = L < \infty \quad \Longrightarrow \quad f(n) = O(g(n)).$$

2 Omega Notation

Let $f, g : \mathbb{N} \to \mathbb{R}_{\geq 0}$, with $g(n) > 0$ for all sufficiently large n. We say that

$$f(n) = \Omega(g(n))$$

if there exist constants $c > 0$ and $n_0 \in \mathbb{N}$ such that

$$0 \leq c\, g(n) \leq f(n) \qquad \text{for all } n \geq n_0.$$

Thus, $g(n)$ is an asymptotic lower bound for $f(n)$. If the following limit exists, then

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = L > 0 \quad \Longrightarrow \quad f(n) = \Omega(g(n)),$$

where L may also be ∞.

3 Theta Notation

Let $f, g : \mathbb{N} \to \mathbb{R}_{\geq 0}$, with $g(n) > 0$ for all sufficiently large n. We say that

$$f(n) = \Theta(g(n))$$

if there exist constants $c_1, c_2 > 0$ and $n_0 \in \mathbb{N}$ such that

$$0 \leq c_1 g(n) \leq f(n) \leq c_2 g(n) \qquad \text{for all } n \geq n_0.$$

Thus, $g(n)$ is an asymptotically tight bound for $f(n)$. If the following limit exists, then

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = L, \qquad 0 < L < \infty, \quad \Longrightarrow \quad f(n) = \Theta(g(n)).$$

4 Karatsuba's Multiplication

Let x and y be two m-digit numbers, where m is even. Splitting each number into two halves gives

$$x = aB^{m/2} + b, \qquad y = cB^{m/2} + d,$$

where B is the base, and a, b, c, d contain at most $m/2$ digits. Their product is

$$xy = acB^m + (ad + bc)B^{m/2} + bd.$$

Karatsuba's method uses only three multiplications:

$$m_1 = ac, \qquad m_2 = bd, \qquad m_3 = (a + b)(c + d).$$

Since $ad + bc = m_3 - m_1 - m_2$, the product is

$$\boxed{xy = m_1 B^m + (m_3 - m_1 - m_2)B^{m/2} + m_2.}$$

5 Strassen's Matrix Multiplication

Let

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \qquad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}.$$

Strassen's method computes $C = AB$ using the following seven multiplications:

$$\begin{aligned} m_1 &= (a_{11} + a_{22})(b_{11} + b_{22}), & m_2 &= (a_{21} + a_{22})b_{11}, \\ m_3 &= a_{11}(b_{12} - b_{22}), & m_4 &= a_{22}(b_{21} - b_{11}), \\ m_5 &= (a_{11} + a_{12})b_{22}, & m_6 &= (a_{21} - a_{11})(b_{11} + b_{12}), \\ m_7 &= (a_{12} - a_{22})(b_{21} + b_{22}). \end{aligned}$$

The elements of the product matrix are

$$\begin{aligned} c_{11} &= m_1 + m_4 - m_5 + m_7, & c_{12} &= m_3 + m_5, \\ c_{21} &= m_2 + m_4, & c_{22} &= m_1 - m_2 + m_3 + m_6. \end{aligned}$$

We have,

$$C = AB = \begin{bmatrix} m_1 + m_4 - m_5 + m_7 & m_3 + m_5 \\ m_2 + m_4 & m_1 - m_2 + m_3 + m_6 \end{bmatrix}.$$